

QCA Fundamentals Certification Quiz

QCA Fundamentals
Fall 2024

December 4, 2024

1 Mathematical Fundamentals (10pts)

Question 1: *Compute the following tensor products:*

$$(a) \begin{bmatrix} e \\ \pi \end{bmatrix} \otimes \begin{bmatrix} i \\ 2 \end{bmatrix}$$

$$(b) |0\rangle \otimes |1\rangle$$

$$(c) |+\rangle \otimes |-\rangle$$

Question 2: *Compute the following operations (hint: some of these don't require any work!):*

$$(a) H |0\rangle$$

$$(b) X |+\rangle$$

$$(c) HX |0\rangle$$

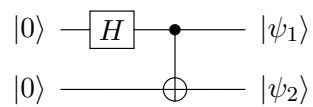
$$(d) XH |0\rangle$$

2 Quantum Information Theory

Question 3: Mark the following as **True** or **False**

- (a) Quantum Operators may be represented as non-square matrices (*True / False*)
- (b) Quantum Operators must be reversible (*True / False*)
- (c) The Measurement operation is reversible (*True / False*)
- (d) According to Born's rule, for $|\psi\rangle = a|0\rangle + b|1\rangle$, $a + b = 1$

Question 4: What is the state of $|\psi_1\rangle$?



3 Entanglement

Question 5: *Which of the following qubits are entangled?*

(a) $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$

(b) $\frac{1}{\sqrt{2}}(|001\rangle + |110\rangle)$

(c) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

(d) $\frac{1}{\sqrt{3}}(|10\rangle + |11\rangle + |00\rangle)$

Question 6: *Which of the four states in question 5 are bell states?*

Question 7: *Draw a circuit that generates an entangled state.*

4 Teleportation

Question 8: *Quantum Teleportation is a purely quantum algorithm. (True / False)*

Question 9: *Why doesn't Quantum Teleportation allow for faster-than-light communication?*

- (a) Quantum Teleportation doesn't help us communicate.
- (b) Quantum Teleportation doesn't rely on entanglement.
- (c) We need to communicate information in a classical manner.
- (d) Quantum Teleportation *does* allow for faster-than-light communication.

Question 10: *Quantum Teleportation allows us to clone a quantum state. (True / False)*

5 Deutsch - Josza

Question 11: *What is the general form of a quantum function?* I.e. if we have a function f that we can apply to a qubit $|x\rangle$, and two qubit registers $|x\rangle, |y\rangle$, fill in the ?:

$$|x\rangle |y\rangle \rightarrow |?\rangle |?\rangle$$

Hint: $\oplus$ is the CNOT operator.

Question 12: *What is the purpose of the Deutsch - Josza algorithm?*

- (a) Factor large numbers into their prime factors.
- (b) Sort an array in $O(\sqrt{n})$ time, faster than a classical computer!
- (c) Simulate the behaviour of a classical system on a quantum computer.
- (d) Calculate whether a given function is balanced or constant.

6 Quantum Hardware

Question 13: *Which of these are valid approaches for representing qubits physically?*

- (a) Trapped-Ion
- (b) Superconducting
- (c) Vibrating Nanotube Oscillator
- (d) Plasma Spin-Wave